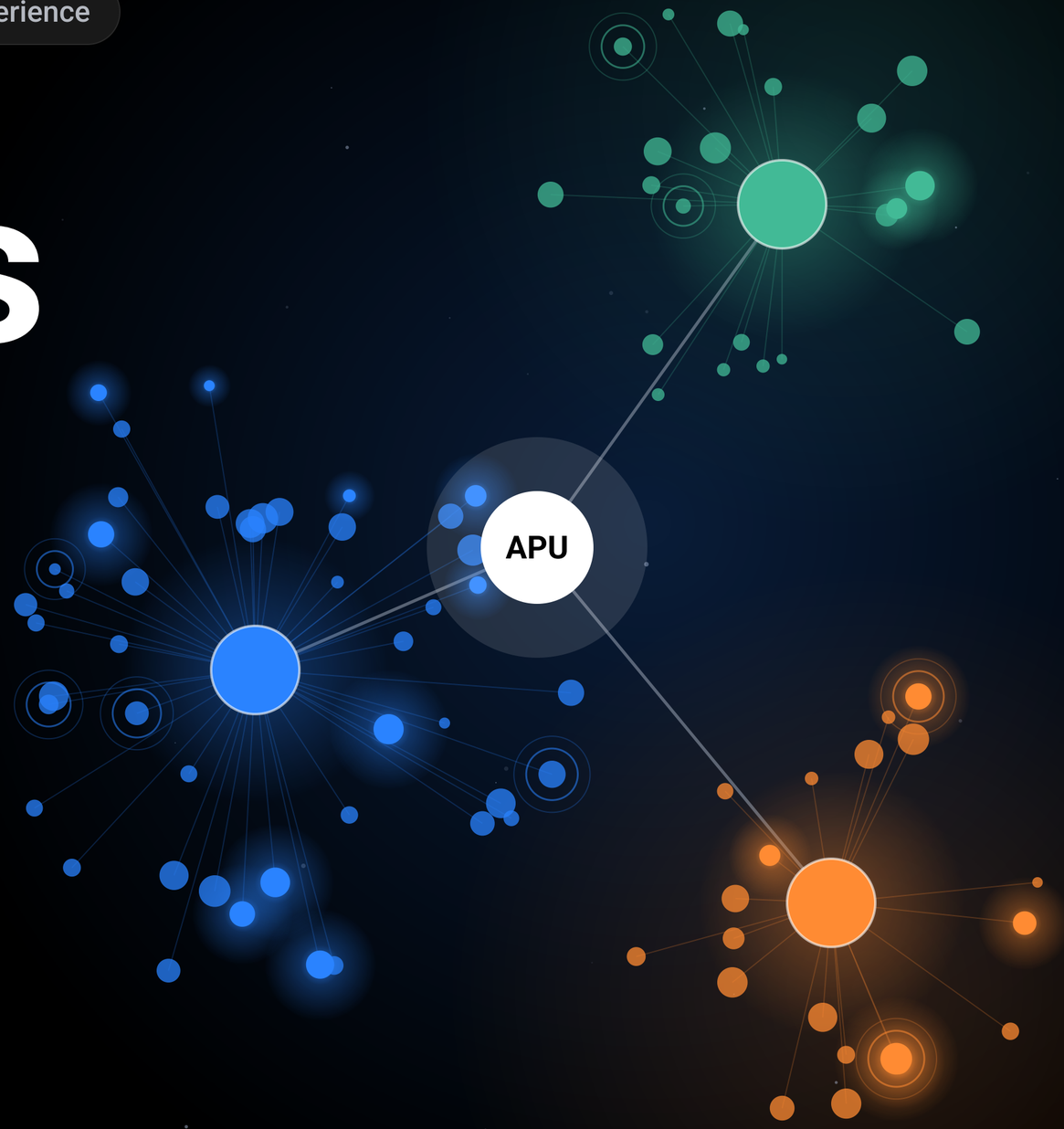
APU Cloud Innovation Hackathon 2026

Track · Smart Campus & Student Experience

APU Campus Graph

Every club at APU. **One living map.**
Just ask.



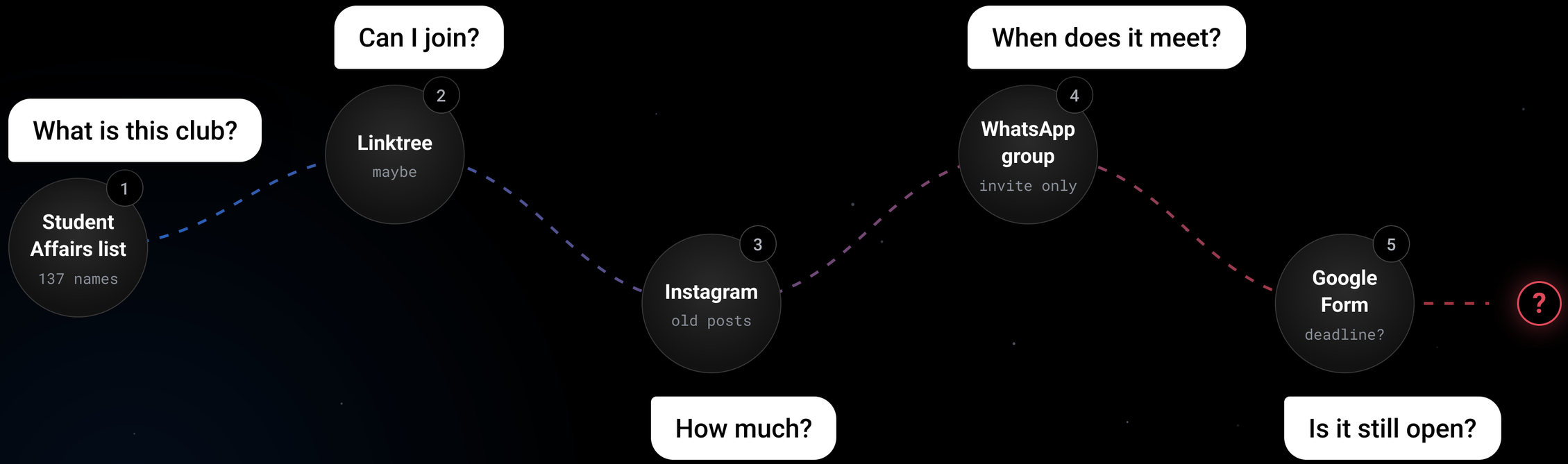
Team Vancomycin

Chang Kai Yang & Joseph Ang

Organised by APU School of Technology with GDG on Campus APU
Sponsors include Google Cloud and NTT DATA

Joining one club takes five tabs of research.

Every answer lives somewhere different.
Some answers aren't online at all.



We read every listing.
All 137 of them.

- 78 societies & SIGs
- 31 sports
- 28 international

0/137

list a **fee**



0/137

have a **join link**



8/137

state **meeting times**



11/137

state **who can join**



The information exists in pieces, never in one place.

Source: studentaffairs.apu.edu.my, scraped 13 Nov 2026

And the campus feels cold.

You can't feel a club before joining.

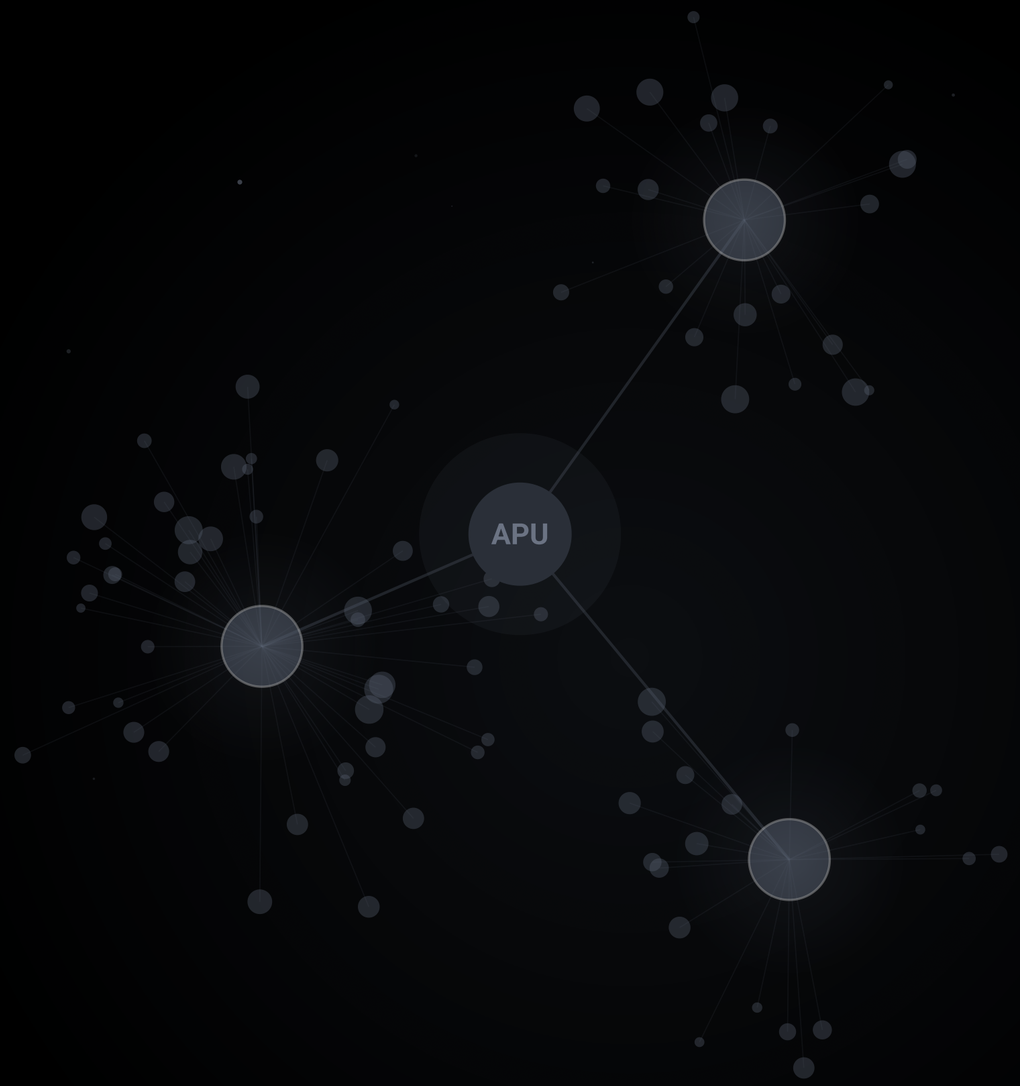
A name on a list says nothing about the people or the vibe.

You don't know who else is going.

Walking in alone is the hardest part.

Nothing nudges you to take the first step.

So the first semester passes, and you never go.



One living map that answers you.

01 · TELLS YOU DIRECTLY

Just ask.

Ask APU, powered by Gemini, gives the answer, not another link.

Can a Year 1 join the AI Club?

Yes, it's open to all APU students. Here's how to join →

02 · SHOWS WHAT'S NEXT

Brighter star, sooner event.

Pulsing rings mean it's happening today.



03 · MAKES YOU WANT TO GO

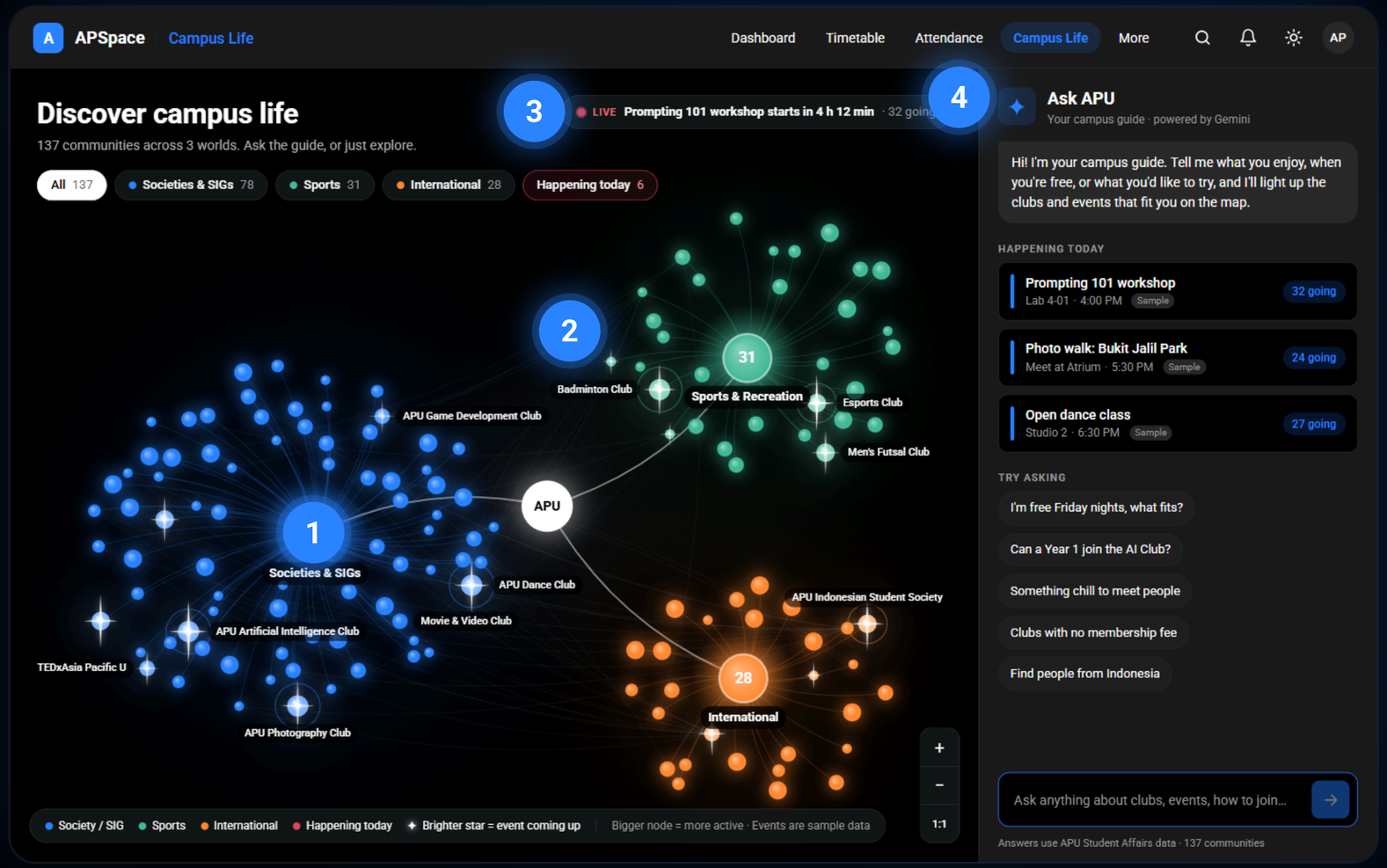
Your first step, made easy.

Small nudges that turn "maybe" into "I'm going".

• Happening today
Go together · 24
✓ Can I join?
Fits your timetable

137 communities.
One glance.

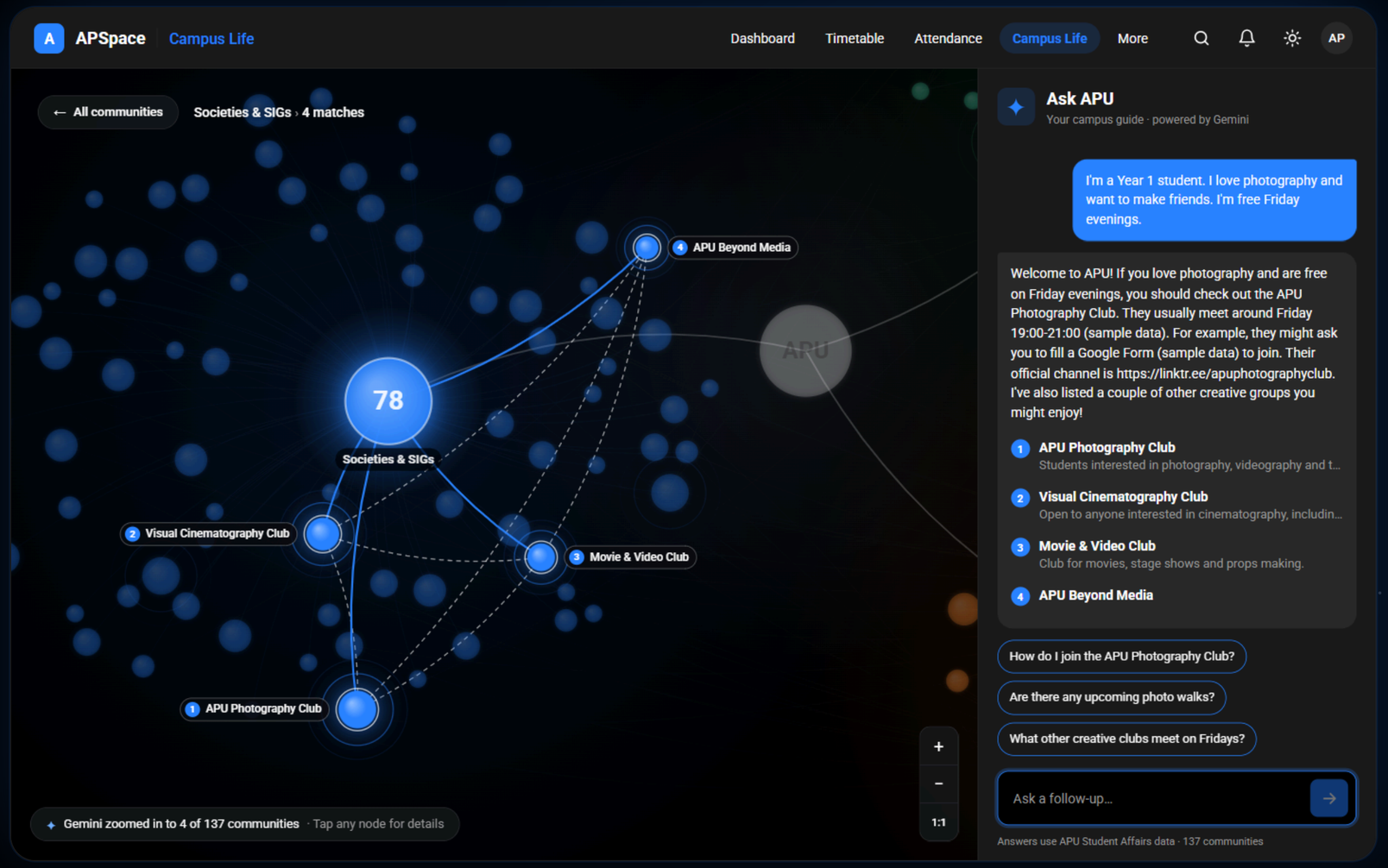
1. **Three worlds**
   Societies, sports and international, around APU.
2. **Pulsing = today**
   Brighter stars have events soon.
3. **Live ticker**
   What's starting soon, and who's going.
4. **Ask APU**
   A guide that lights up the map.



↑ Brighter star = event coming up

Ask, and the map answers.

1. **Ask in your own words**
   "Year 1, love photography, free on Friday nights."
2. **Gemini answers from the catalog**
   Only real APU communities. It never invents a club.
3. **The map zooms in**
   4 matches light up, and the best one opens its card.



Everything on one card.

Can I join?

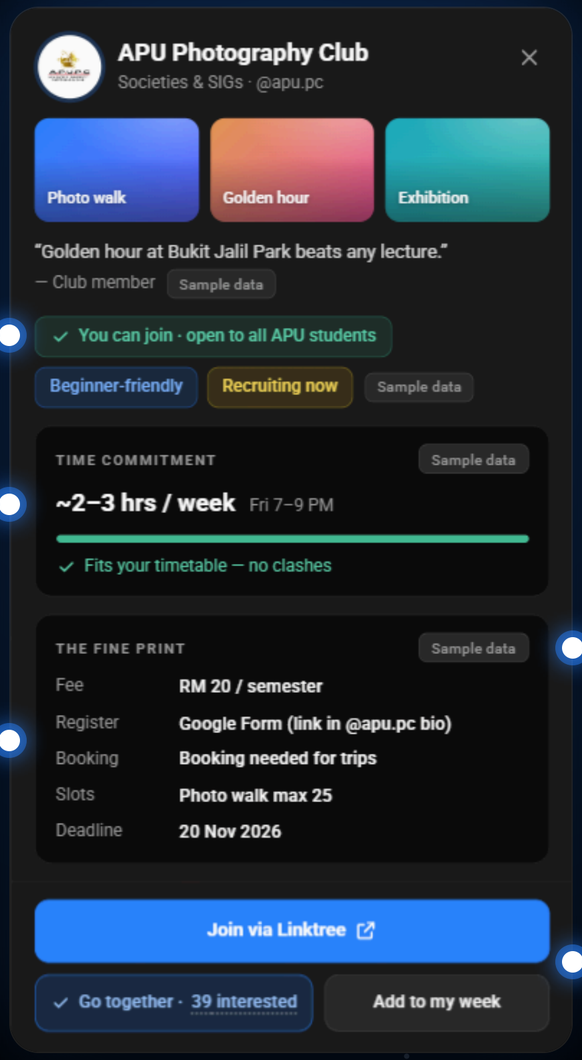
A clear yes, and who it's open to.

Time and timetable fit

Hours per week, checked against your classes.

The fine print

Fee, register, booking, slots, deadline.



Values not published by clubs are shown with a Sample data tag.

Sample data clearly labelled until clubs publish real details

Join · Go together · Add to my week

One tap from "maybe" to "I'm in".

A Gemini agent with tools.
Checked before you see it.

BROWSER

React + Vite

Single-page app. Living graph and Ask APU chat.

AGENT LOOP · EXPRESS

Plan → tools → check

Gemini picks tools, the server runs them on real data, a validator checks the answer and sends mistakes back once.

GEMINI FUNCTION CALLING

search_clubs · get_club · events

Answer = JSON actions: focus, open card

CATALOG

137 real communities

Scraped from Student Affairs

- **Self-checking** — Made-up clubs, sample fees stated as fact and hijacked voices are caught and fixed.
- **Never dies** — Retry on 503, warm-up on load, then an offline matcher.
- **Cloud Run ready** — Dockerised for Cloud Run. Live demo hosted on Render.
- **Honest data** — Sample values are clearly labelled as sample data.

09 — IMPACT & NEXT STEPS

From a list of names to a first step.

IMPACT

Fewer hours researching

More first-time joiners

Clubs reach new members

ROADMAP

- **Gemini reads club posters → fine print fills itself**
  Paste a caption or poster; details fill in.
- **Student Affairs feed & club admin**
  Clubs keep their own card up to date.
- **APSpace timetable sync**
  Real "fits your timetable" for every student.
- **Real event calendar**
  Every star's brightness comes from live events.
- **Attendance → recommendations**
  Suggest the next club after the first one.

LIVE DEMO apu-campus-graph.onrender.com

CODE github.com/youamar/apu-campus-graph

VIDEO in repo /video

Every club at APU. One living map. Just ask.